Requirement Analysis Phase

Introduction

The Requirement Analysis Phase focuses on identifying and defining the functional and non-functional requirements of the Automated Ticket Assignment System. This phase ensures that the system clearly addresses the problems identified in the ideation phase and provides a structured solution using the ServiceNow platform.

In this project, the requirement analysis helps determine how tickets should be created, how they should be categorized, and how they will be automatically assigned to the appropriate support group. By analyzing these requirements, the system can be designed to ensure efficient ticket handling, faster resolution times, and better management of support operations.

The ServiceNow platform provides various features such as user management, role-based access control, table creation, and workflow automation, which are used to meet the project requirements. These features help in creating a structured and secure environment for managing support tickets.

Functional Requirements

Functional requirements describe the specific operations that the system must perform.

The Automated Ticket Assignment System should provide the following functionalities:

• The system should allow users to create support tickets with issue details.
• The system should categorize tickets based on predefined issue types.
• The system should automatically assign tickets to the appropriate support group based on the issue type.
• The system should allow administrators to manage users, groups, and roles.
• The system should allow support teams to view and update ticket details.
• The system should maintain records of all tickets for tracking and reporting purposes.

These functions ensure that the ticket management process becomes efficient and automated.

Non-Functional Requirements

Non-functional requirements describe the system characteristics such as performance, security, and usability.

• **Performance:** The system should automatically assign tickets without delay when a new ticket is created.

• **Security:** Access to ticket data should be controlled using roles and access control rules.
• **Usability:** The system interface should be simple and easy for users and support staff to use.
• **Reliability:** The system should ensure accurate ticket routing and minimal errors.
• **Scalability:** The system should handle increasing numbers of support requests.

These requirements ensure that the system operates effectively and securely.

Roles in the System

Different users interact with the system based on their responsibilities.

Administrator

- Manages system configuration.
- Creates users, groups, roles, and tables.
- Defines automation rules and workflows.

Support Team Members

- Receive tickets assigned to their group.
- Analyze the problem and provide solutions.
- Update ticket status after resolving the issue.

End Users

- Report issues by creating tickets.
- Provide necessary information related to the problem.

Role-based access control ensures that each user can only perform the actions permitted for their role.

Database Tables

Database tables are used to store the information related to support tickets and users.

The main table created for this project is:

Operations Related Table

This table stores all the support ticket information.

Important fields in this table include:

• Issue
• Description
• Assigned To
• Status
• Ticket Raised Date
• Priority

The Issue field contains predefined options such as:

• Unable to login to platform
• 404 error
• Regarding certificates
• Regarding user expired

These choices help standardize ticket creation and make it easier to apply automated assignment rules.

System Components

The system consists of several components that work together to automate ticket assignment.

Users
Users represent individuals who interact with the system. In this project, different users are created for testing the support process.

Groups
Groups represent teams responsible for handling specific types of issues. Examples include the Platform Support Group and the Certificate Support Group.

Roles
Roles define what actions users can perform within the system. Roles ensure that only authorized users can access or modify ticket data.

Access Control Rules (ACL)
ACLs restrict access to tables and fields to ensure data security.

Flows
Flows automate the ticket assignment process. When a ticket is created with a specific issue type, the system automatically assigns it to the correct support group.